FIRESHIELD - FIRE INSURANCE MANAGEMENT SYSTEM

SOFTWARE REQUIREMENTS SPECIFICATION

ISO/IEC/IEEE 29148:2018 Compliant | Version 1.0 – Final

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Table of Contents

1. Introduction, Scope & Overview 3
2. Stakeholder Analysis & Use Cases 5
3. Functional Requirements – Customer Module 7
4. Functional Requirements – Staff & Administrator Modules 9
5. Premium & Policy Calculation Engine 11
6. Policy Lifecycle – State Machine Specification 12
7. Claim Lifecycle – State Machine Specification 14
8. Database Schema & Relational Design 16
9. Security Architecture & API Contracts 19
10. Frontend Architecture – Angular Design 22
11. Audit Log Dictionary & Invoice Generation 24
12. Testing, QA & Non-Functional Requirements 26
13. Deployment, Scalability & Future Roadmap 28
14. Glossary, Acronyms & Revision History 30

Chapter 1: Introduction, Scope & Overview

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, technical, and non-functional requirements for the FireShield - Fire Insurance Management System (FSMS). It serves as the authoritative baseline between stakeholders and the development team, structured in accordance with ISO/IEC/IEEE 29148:2018.

1.2 Product Description

FSMS is a full-stack web application designed to manage the complete fire insurance lifecycle — from policy subscription and premium calculation through to claim adjudication, property inspection, and settlement — across five role-segregated portals: Public, Customer, Surveyor, Underwriter, and Administrator.

Architecture Overview

Architecture Layer	Responsibility	Technology
Domain Layer	Core entities, enums, repository interfaces; zero external dependencies	Java POJOs / TypeScript Models
Application Layer	Business logic, services, DTOs, use-case orchestration	Spring Boot Services / Angular Services
Infrastructure Layer	JPA Repositories, PDF generation, file storage	MySQL/PostgreSQL, iText/JasperReports
API Layer	Controllers, JWT middleware, CORS, global error handling	Spring Boot REST / Express.js

1.3 Technology Stack

Component	Technology	Version
Frontend Runtime	Angular (Standalone Components)	Angular 15+
Frontend Language	TypeScript, HTML, CSS	TypeScript 5+
Backend Runtime	Spring Boot (Java) or Node.js/Express	Spring Boot 3 / Node 18+
ORM / Data Access	JPA / Hibernate or TypeORM	Latest Stable
Database	MySQL or PostgreSQL	8.0+ / 15+
Authentication	JWT Bearer / BCrypt	RFC 7519 / BCrypt v2
PDF Generation	iText / JasperReports	v7+ / v6+
API Documentation	Swagger / OpenAPI	3.0
Unit Testing	JUnit + Mockito / Jest	JUnit 5 / Jest 29

1.4 System Constraints

- All monetary fields must use DECIMAL(18,2) to prevent floating-point rounding errors.
- JWT tokens carry a mandatory 2-hour sliding expiration; refresh tokens are scoped per session.

- Document uploads restricted to PDF, JPG, PNG; maximum 10 MB per file.
- Claim deduction and coverage update operations must execute within ACID-compliant transactions.
- Role segregation must prevent cross-role portal access under all session configurations.
- CORS policy restricts API access to explicitly whitelisted Angular origin URLs.
- All PII fields (FullName, Email, Address) must never appear in application log output.

1.5 References

- ISO/IEC/IEEE 29148:2018 – Systems and Software Engineering Requirements
- OWASP Web Application Security Guidelines v4.2
- Angular 15+ Official Documentation – Google
- Spring Boot 3 / Node.js 18 Official Documentation
- OpenAPI / Swagger Specification v3.0

Chapter 2: Stakeholder Analysis & Use Cases

2.1 Stakeholder Registry

Stakeholder	Role	Primary Interests
Customer (Policyholder)	End user	Policy status, coverage balance, claim outcomes, premium invoices, inspection reports
Surveyor	Field inspector	Assigned inspection queue, site visit scheduling, report submission
Underwriter	Risk assessor	Policy review queue, risk rating, premium approval/rejection, coverage limits
System Administrator	Super-user	Staff management, plan configuration, global audit access, revenue KPIs, dashboard analytics
Compliance Officer	External auditor	Audit trail completeness, RBAC enforcement, data protection, regulatory reporting
Public / Anonymous	Prospective customer	Premium quotation, plan information, registration

2.2 Use Case Summary

UC ID	Use Case	Actor	Precondition
UC-01	Calculate Insurance Premium Quote	Public / Customer	Plan templates exist
UC-02	Register & Submit Policy Application	Customer (Registered)	Quote generated; user authenticated
UC-03	Process Payment & Activate Policy	Customer (Registered)	Policy approved by Underwriter
UC-04	Submit FireShield Insurance Claim	Customer (Registered)	Active policy; sufficient coverage
UC-05	Assign Surveyor to Inspection	Administrator	Claim submitted; surveyor available
UC-06	Conduct Property Inspection & Submit Report	Surveyor	Inspection assigned; site accessible
UC-07	Underwrite & Adjudicate Claim	Underwriter	Inspection report received; claim PendingReview
UC-08	Approve / Reject Policy Application	Underwriter	All documents Verified; risk assessment complete
UC-09	Onboard New Staff Member	Administrator	Admin authenticated session
UC-10	Create / Edit Plan Template	Administrator	Admin authenticated session
UC-11	Generate & Store PDF Invoice	System (Automated)	Payment transaction confirmed
UC-12	View Admin Analytics Dashboard	Administrator	Authenticated admin session

2.3 Policy Application Flow (UC-02) – End-to-End

The policy application lifecycle is the most complex multi-actor flow in the system:

- Customer selects a plan tier (Basic/Standard/Premium) and enters property details; PremiumService calculates quote.
- Customer submits application — PolicySubscription record created with Status = PendingReview.
- Customer uploads mandatory documents (Property Title, ID Proof, Property Photos) to DocumentVault.
- Admin assigns the application to an Underwriter based on risk class and geography.
- Underwriter reviews each document individually, setting status to Verified or Rejected.
- Document Gate: if all Verified → policy Approved; if any Rejected → auto-Rejected.
- On approval, customer completes payment; system generates PDF invoice, activates policy (Status = Active).

2.4 Claim Adjudication Flow (UC-04 to UC-07) – End-to-End

- Customer submits a fire damage claim with ClaimAmount, Reason, and supporting documents.
- Admin assigns a Surveyor to conduct on-site property inspection.
- Surveyor submits inspection report; claim transitions to InspectionComplete.
- Underwriter reviews claim, inspection report, and supporting documents.
- Deduction Gate: if ClaimAmount exceeds CoverageAmount → auto-reject with HTTP 400.
- On approval, CoverageAmount is atomically decremented; PDF settlement notice generated.

Chapter 3: Functional Requirements – Customer Module

3.1 Authentication & Registration

ID	Requirement	Priority
FR-AUTH-01	Register new users with BCrypt-hashed passwords; Customer role assigned at point of creation.	Must Have
FR-AUTH-02	Return signed JWT on login containing sub (UserID), email, and role claims.	Must Have
FR-AUTH-03	JWT tokens expire after 2 hours; expired tokens return HTTP 401.	Must Have
FR-AUTH-04	Reject login for accounts marked IsActive = false without exposing reason.	Should Have
FR-AUTH-05	Support password reset via email OTP or secure token link.	Should Have

3.2 Premium Quotation

ID	Requirement	Priority
FR-QUOT-01	Calculate monthly premium based on property value, location risk zone, and plan tier multiplier.	Must Have
FR-QUOT-02	Apply tier multipliers: Basic 1.0x, Standard 1.3x, Premium 1.6x.	Must Have
FR-QUOT-03	CoverageAmount = MonthlyPremium x 120 x TierMultiplier.	Must Have
FR-QUOT-04	Persist quote as PolicySubscription with Status = PendingReview.	Must Have
FR-QUOT-05	Quotation calculation accessible without authentication (anonymous/public access).	Must Have

3.3 Document Upload

ID	Requirement	Priority
FR-DOC-01	Accept document types: Property Title, ID Proof, Property Photos, Fire Report, Medical Bill, Invoice.	Must Have
FR-DOC-02	Enforce file type restriction: .pdf, .jpg, .png only.	Must Have
FR-DOC-03	Enforce maximum file size: 10 MB per document.	Must Have
FR-DOC-04	Documents stored in DocumentVault with RelatedEntityType linking to Policy or Claim.	Must Have
FR-DOC-05	Document status defaults to Pending on upload.	Must Have

3.4 Claims Submission

ID	Requirement	Priority
FR-CLM-01	Allow customers to submit fire damage claims against active policies with sufficient coverage.	Must Have

FR-CLM-02	Claim submission requires: ClaimAmount, DamageDescription, IncidentDate, and at least one supporting document.	Must Have
FR-CLM-03	Display claim history with color-coded status badges (Submitted / Inspecting / Approved / Rejected / Settled).	Must Have
FR-CLM-04	Customers cannot view or modify other customers' claims under any session configuration.	Must Have
FR-CLM-05	Customer receives in-app notification on claim status transitions.	Should Have

3.5 Customer Policy Dashboard

ID	Requirement	Priority
FR-DASH-01	Dashboard displays all active, pending, and expired policies with real-time CoverageAmount.	Must Have
FR-DASH-02	Customer can download PDF invoice for each confirmed payment.	Must Have
FR-DASH-03	Customer can view inspection status and surveyor report summary for open claims.	Should Have

Chapter 4: Functional Requirements – Staff & Administrator Modules

4.1 Surveyor Module

ID	Requirement	Priority
FR-SRV-01	Surveyor portal displays only inspection assignments allocated by the Administrator.	Must Have
FR-SRV-02	Surveyor can update inspection status: Assigned to InProgress to Completed.	Must Have
FR-SRV-03	Surveyor submits a structured inspection report with damage assessment, photos, and estimated loss value.	Must Have
FR-SRV-04	Inspection report submission transitions the associated claim to InspectionComplete.	Must Have
FR-SRV-05	Surveyor views historical assignments and completed reports.	Should Have

4.2 Underwriter Module

ID	Requirement	Priority
FR-UND-01	Underwriter inbox shows only policies and claims assigned by the Administrator.	Must Have
FR-UND-02	Underwriter can set each document status individually: Pending to Verified or Rejected.	Must Have
FR-UND-03	Document Gate: policy approval proceeds only if ALL documents are Verified.	Must Have
FR-UND-04	Auto-reject policy application if ANY document is set to Rejected.	Must Have
FR-UND-05	Underwriter adjudicates claims; Deduction Gate blocks approvals where ClaimAmount > CoverageAmount.	Must Have
FR-UND-06	On claim approval, atomically deduct ClaimAmount from policy's CoverageAmount.	Must Have
FR-UND-07	Underwriter can add an internal adjudication note to each claim decision.	Should Have

4.3 Administrator Module

ID	Requirement	Priority
FR-ADM-01	Admin is the only role authorized to create Staff (Surveyor, Underwriter) accounts.	Must Have
FR-ADM-02	Admin creates Plan Templates with PlanName, BasePremium, TierMultiplier, and RiskZone.	Must Have
FR-ADM-03	Admin assigns/reassigns Underwriters to policy queues and Surveyors to inspection requests.	Must Have

FR-ADM-04	Admin has unfiltered read access to the complete AuditActionLog.	Must Have
FR-ADM-05	Admin dashboard displays: Total Active Policies, MRR, Claims Approval Rate, Avg Claim Amount, Avg Settlement Time, Active Policy Ratio, Pending Inspections, recent claims, and pending subscriptions.	Must Have
FR-ADM-06	Admin soft-deletes staff/customer accounts via IsActive flag (no hard deletes).	Should Have
FR-ADM-07	Admin can view, filter, and export audit logs by date range, actor, and action type.	Should Have
FR-ADM-08	Admin dashboard supports visual analytics: pie chart (claim status), bar chart (policy trends), line chart (monthly revenue).	Nice to Have

Chapter 5: Premium & Policy Calculation Engine

5.1 Engine Architecture

The Premium Calculation Engine (PremiumService) is a stateless service in the Application Layer, ensuring a fresh instance per request with no state contamination between concurrent calculations. It consumes admin-configured plan templates for base premium values and risk zone adjustments.

5.2 Calculation Formulae

Monthly Premium: Base Premium x (1 + (PropertyRiskScore / 10) x 0.10) x TierMultiplier

Coverage Amount: Monthly Premium x 120 x TierMultiplier

Agent Commission: Monthly Premium x 0.05 (persisted on policy activation)

5.3 Tier & Boundary Value Reference

Plan	Tier Mult.	$100K Property	$200K Property	$500K Property	Coverage
Basic	1.0x	$550/mo	$700/mo	$1,000/mo	Prem x120
Standard	1.3x	$715/mo	$910/mo	$1,300/mo	Prem x120x1.3
Premium	1.6x	$880/mo	$1,120/mo	$1,600/mo	Prem x120x1.6

5.4 Dashboard Analytics (DashboardService)

The Admin Dashboard aggregates cross-entity metrics using optimized read-only queries:

KPI	Source Tables	Calculation Logic
Total Active Policies	PolicySubscriptions	COUNT where Status = Active
Monthly Recurring Revenue (MRR)	PolicySubscriptions	SUM(MonthlyPremium) from all active policies
Claims Approval Rate	Claims	COUNT(Approved) / COUNT(Total) x 100
Average Claim Amount	Claims	SUM(ClaimAmount) / COUNT(Approved Claims)
Average Settlement Time	Claims	AVG(SettledAt - SubmittedAt) in days
Active Policy Ratio	PolicySubscriptions	COUNT(Active) / COUNT(All) x 100
Pending Inspections	Inspections	COUNT where Status = Assigned OR InProgress
Claim Payout Liquidity	Claims	SUM(ClaimAmount) where Status = Approved

Chapter 6: Policy Lifecycle – State Machine Specification

6.1 State Transition Matrix

The Policy Subscription follows a six-state machine with enforced transitions. Invalid transitions are rejected at the service layer; every valid transition is recorded in the AuditActionLog.

Current State	Triggering Event	Next State	Actor	Gate Condition
[None]	Customer submits quote application	PendingReview	Customer	Authenticated; quote calculated
PendingReview	Admin assigns to Underwriter	PendingReview*	Admin	Underwriter exists for plan
PendingReview	Underwriter approves all documents	Approved	Underwriter	ALL docs Verified; zero Rejected
PendingReview	Any document rejected	Rejected	Underwriter / System	One or more docs Rejected
Approved	Customer completes payment	Active	Customer + System	Payment gateway confirmed
Active	Expiry date reached	Expired	System (Scheduled)	ExpiryDate < current UTC
Active	Customer requests cancellation	Cancelled	Customer / Admin	No pending claims
Rejected	—	[Terminal]	N/A	No further transitions

6.2 State Descriptions

Draft / Quotation

Prospect calculates premium (anonymous or authenticated). No persistent DB record yet. No obligations on either party.

PendingReview (Status = 0)

Application formally submitted with documents uploaded. Awaiting Admin assignment and Underwriter document review. All documents default to Pending status.

Approved (Status = 1)

All documents verified by Underwriter. Awaiting customer payment. Policy not yet active; coverage not yet available.

Active (IsPaid = true)

Payment confirmed. PDF invoice generated and stored in DocumentVault. Policy live on customer dashboard with real-time CoverageAmount display.

Rejected (Status = 2)

Terminal state. Underwriter found forged or invalid documents. No further transitions permitted.

Expired / Cancelled

Policy term has ended or customer has cancelled. Historical record preserved; no coverage available.

6.3 Document Gate Logic (UnderwriterController.VerifyPolicy)

This is the most critical business logic guard in the policy workflow:

- Retrieve all DocumentVault entries linked to the target PolicySubscriptionId.
- If any entry has Status = Rejected, immediately set application to Rejected (Status=2) and write audit log.
- If any entry has Status = Pending, return HTTP 400: 'All documents must be reviewed before approval.'
- If all entries are Verified, approve application; set Status = Approved (1).
- Write AuditActionLog: ActionType = PolicyVerification, OldValue, NewValue, PerformedByUserId, Timestamp.

Chapter 7: Claim Lifecycle – State Machine Specification

7.1 State Transition Matrix

Current State	Triggering Event	Next State	Actor	Gate Condition
[None]	Customer submits fire damage claim	Submitted	Customer	Active policy; amount > 0; docs uploaded
Submitted	Admin assigns Surveyor	Inspecting	Administrator	Surveyor available
Inspecting	Surveyor submits inspection report	InspectionComplete	Surveyor	Report with photos and loss value
InspectionComplete	Underwriter approves; coverage sufficient	Approved	Underwriter + System	ClaimAmount <= CoverageAmount
InspectionComplete	Underwriter rejects or coverage insufficient	Rejected	Underwriter / System	ClaimAmount > CoverageAmount OR fraud
Approved	Settlement processed	Settled	System (Automated)	Payment disbursed; notice generated
Settled	—	[Terminal]	N/A	Coverage deducted; audit logged
Rejected	—	[Terminal]	N/A	No deduction; customer may appeal

7.2 Deduction Gate Logic (UnderwriterController.DecideClaim)

The DecideClaim endpoint is a safety-critical operation modifying the policy's remaining CoverageAmount. The following sequence executes upon underwriter approval:

- Retrieve the Claim record with its linked PolicySubscription parent.
- Coverage check: if Claim.ClaimAmount > Policy.CoverageAmount, return HTTP 400: 'Insufficient coverage balance.' Abort.
- Begin database transaction (ACID-compliant).
- Set Claim.Status = Approved.
- Compute: Policy.CoverageAmount = Policy.CoverageAmount minus Claim.ClaimAmount.
- Write AuditActionLog: ActionType = ClaimApprovalDeduction; OldValue = previous coverage; NewValue = updated coverage.
- Commit transaction atomically. If audit log write fails, entire transaction rolls back.
- Generate PDF settlement notice and store in DocumentVault.
- Return HTTP 200 with updated claim and coverage data.

7.3 Inspection Workflow

Upon a claim reaching Submitted status:

- Admin reviews claim and selects an available Surveyor based on location and availability.
- Surveyor receives assignment notification; status transitions to Inspecting.

- Surveyor conducts on-site visit, captures photos, assesses structural damage, and estimates loss value.
- Surveyor submits structured report; status transitions to InspectionComplete.
- Report becomes available to Underwriter for final adjudication.

7.4 Customer Claim Dashboard

The GetMyClaims endpoint returns all claims associated with a customer's policies in reverse chronological order, showing ClaimAmount, Status (color-coded badge), submission date, DamageDescription, and inspection report summary. The customer's live dashboard also displays the remaining CoverageAmount in real time after each approved claim deduction.

Chapter 8: Database Schema & Relational Design

8.1 Relational Design Principles

The schema is managed by ORM code-first migrations against MySQL/PostgreSQL. All parent-child relationships use cascade RESTRICT to prevent accidental deletion of customer or claim history. All monetary columns are explicitly configured to DECIMAL(18,2).

8.2 Entity Data Dictionary

Table: Users

Column	Type / Constraints	Description
id	INT, PK, AUTO_INCREMENT	Auto-incremented primary key
fullName	VARCHAR(200), NOT NULL	Legal full name of the user
email	VARCHAR(256), NOT NULL, UNIQUE	Login email address
passwordHash	VARCHAR(MAX), NOT NULL	BCrypt v2 hashed password
role	VARCHAR(50), NOT NULL	Enum: Customer, Surveyor, Underwriter, Admin
isActive	BOOLEAN, DEFAULT true	Soft delete flag
createdAt	TIMESTAMP, NOT NULL	Account creation timestamp (UTC)

Table: PolicySubscriptions

Column	Type / Constraints	Description
id	INT, PK, AUTO_INCREMENT	Auto-incremented primary key
userId	INT, FK -> Users.id RESTRICT	Owning customer
subscriptionReference	VARCHAR(20), UNIQUE	System-generated ref (POL-XXXX)
selectedPlanName	VARCHAR(100), NOT NULL	Basic / Standard / Premium
monthlyPremium	DECIMAL(18,2), NOT NULL	Risk-adjusted monthly premium
coverageAmount	DECIMAL(18,2), NOT NULL	Remaining coverage balance
status	INT, NOT NULL, DEFAULT 0	0=Pending, 1=Approved, 2=Rejected
isPaid	BOOLEAN, DEFAULT false	Payment confirmation flag
underwriterId	INT, FK -> Users.id, NULLABLE	Assigned underwriter reference
expiryDate	TIMESTAMP, NULLABLE	Policy expiration (UTC)

Table: Claims

Column	Type / Constraints	Description
id	INT, PK, AUTO_INCREMENT	Auto-incremented primary key

userId	INT, FK -> Users.id RESTRICT	Customer who filed the claim
policySubscriptionId	INT, FK -> PolicySubscriptions.id RESTRICT	Linked active policy
claimAmount	DECIMAL(18,2), NOT NULL	Requested monetary amount
damageDescription	TEXT, NOT NULL	Fire damage description text
incidentDate	DATE, NOT NULL	Date of the fire incident
status	VARCHAR(50), NOT NULL	Submitted / Inspecting / InspectionComplete / Approved / Rejected / Settled
submittedAt	TIMESTAMP, NOT NULL	UTC submission timestamp
settledAt	TIMESTAMP, NULLABLE	UTC settlement timestamp

Table: Inspections

Column	Type / Constraints	Description
id	INT, PK, AUTO_INCREMENT	Auto-incremented primary key
claimId	INT, FK -> Claims.id RESTRICT	Associated claim
surveyorId	INT, FK -> Users.id RESTRICT	Assigned surveyor
status	VARCHAR(50), NOT NULL	Assigned / InProgress / Completed
estimatedLossValue	DECIMAL(18,2), NULLABLE	Surveyor's damage estimate
reportNotes	TEXT, NULLABLE	Structured inspection report text
inspectedAt	TIMESTAMP, NULLABLE	UTC inspection completion time

Table: DocumentVault

Column	Type / Constraints	Description
id	INT, PK, AUTO_INCREMENT	Auto-incremented primary key
fileName	VARCHAR(300), NOT NULL	Original file name
fileData	BLOB / BYTEA, NOT NULL	Binary file content
contentType	VARCHAR(100), NOT NULL	MIME type
documentType	VARCHAR(50), NOT NULL	PropertyTitle / IDProof / PropertyPhoto / FireReport / Invoice
status	VARCHAR(50), DEFAULT 'Pending'	Pending / Verified / Rejected
relatedEntityType	VARCHAR(50), NOT NULL	Discriminator: Policy or Claim
relatedEntityId	INT, NOT NULL	FK to parent entity

Table: AuditActionLog

Column	Type / Constraints	Description

id	INT, PK, AUTO_INCREMENT	Log entry identifier (append-only)
actionType	VARCHAR(100), NOT NULL	Event category (see Audit Dictionary, Ch. 11)
relatedEntityId	INT, NOT NULL	PK of the modified entity
oldValue	TEXT	Serialized pre-action state
newValue	TEXT	Serialized post-action state
performedByUserId	INT, FK -> Users.id	Actor's user ID
timestamp	TIMESTAMP (UTC), NOT NULL	Precise UTC event timestamp

Chapter 9: Security Architecture & API Contracts

9.1 JWT Authentication Flow

FireShield uses stateless JWT-based authentication aligned with OWASP Top 10 guidelines, with specific mitigations for Broken Access Control (A01), Cryptographic Failures (A02), and Injection (A03).

- User submits credentials to POST /auth/login; BCrypt constant-time comparison prevents timing attacks.
- On success, JWT is constructed with claims: sub (UserID), email, role; signed with HMAC-SHA256 (256-bit symmetric key).
- Token returned with 2-hour expiry; Angular's auth.interceptor.ts injects it as Bearer header on every request.
- Server Authentication Middleware validates signature, issuer, audience, and expiry per request.
- Authorization Middleware enforces role attributes; returns HTTP 403 on privilege mismatch.
- Angular role.guard.ts (CanActivate) verifies session token and role before any component initializes.

9.2 RBAC Permission Matrix

Endpoint / Feature	Customer	Surveyor	Underwriter	Admin	Public
POST /auth/register	Self	No	No	No	Yes
POST /auth/addstaff	No	No	No	Yes	No
GET /quote/calculate	Yes	Yes	Yes	Yes	Yes
GET /policy/dashboard	Own	No	No	Yes	No
GET /underwriter/pending-policies	No	No	Assigned	Yes	No
PATCH /underwriter/verify-policy	No	No	Assigned	No	No
GET /claims/pending	No	No	Assigned	Yes	No
PATCH /claims/decide	No	No	Assigned	No	No
GET /inspections/my-queue	No	Own	No	Yes	No
PATCH /inspections/submit-report	No	Assigned	No	No	No
GET /admin/audit-log	No	No	No	All	No
POST /policy/create-template	No	No	No	Yes	No
GET /admin/dashboard	No	No	No	Yes	No

9.3 Data Protection Standards

- All API communication enforced over HTTPS/TLS 1.2+; plain HTTP disallowed in production.
- BCrypt work factor 12 provides resistance to GPU-accelerated brute-force attacks.
- JWT stateless design enables horizontal scaling without session synchronization.
- CORS policy restricts API access to explicitly whitelisted Angular origin URLs.
- PII fields (FullName, Email, Address) never written to application logs.
- File uploads are scanned for MIME type validation server-side; stored as binary blobs.

9.4 Core API Endpoint Catalogue

Method	Endpoint	Role	Description
POST	/api/auth/register	Public	Register new customer account
POST	/api/auth/login	Public	Authenticate and receive JWT
POST	/api/auth/addstaff	Admin	Create Surveyor or Underwriter account
GET	/api/quote/calculate	Public	Calculate premium (anonymous)
POST	/api/quote/request	Customer	Submit and persist quote application
GET	/api/policy/plans	Public	List available plan templates
GET	/api/policy/dashboard	Customer	Customer policy dashboard
POST	/api/policy/create	Admin	Create new plan template
PATCH	/api/policy/assign-underwriter/{id}	Admin	Assign underwriter to application
GET	/api/underwriter/pending-policies	Underwriter	Verification inbox
PATCH	/api/underwriter/verify-policy/{id}	Underwriter	Approve or reject policy
GET	/api/claims/pending	Underwriter	Claim settlement queue
PATCH	/api/claims/decide/{id}	Underwriter	Approve or reject claim
POST	/api/claims/submit	Customer	Submit new fire damage claim
GET	/api/claims/my-claims	Customer	View own claim history
POST	/api/inspections/assign	Admin	Assign surveyor to inspection
GET	/api/inspections/my-queue	Surveyor	Surveyor's inspection assignments
PATCH	/api/inspections/submit-report/{id}	Surveyor	Submit inspection report
POST	/api/payment/process	Customer	Process payment; activate policy
GET	/api/admin/dashboard	Admin	Aggregate KPI analytics
GET	/api/admin/audit-log	Admin	Full audit log access

Chapter 10: Frontend Architecture – Angular Design

10.1 Module & Component Structure

The Angular frontend uses a feature-module architecture with lazy loading for all role-specific sections, minimizing initial bundle size. The Core Module loads eagerly and provides application-wide services and guards.

Module	Lazy?	Key Components	Route
Core Module	No	Navbar, Sidebar, AuthGuard, RoleGuard, JWTInterceptor, LoadingInterceptor	/
Auth Module	Yes	LoginComponent, RegisterComponent	/auth
Customer Module	Yes	DashboardComponent, QuoteWizard, DocumentUpload, ClaimSubmit, PolicyHistory	/customer
Surveyor Module	Yes	InspectionQueueComponent, InspectionDetailComponent, ReportSubmitComponent	/surveyor
Underwriter Module	Yes	PendingPoliciesComponent, DocumentReviewComponent, ClaimQueueComponent, ClaimDetailComponent	/underwriter
Admin Module	Yes	StaffManagementComponent, PlanConfigComponent, AuditDashboard, AnalyticsDashboard, CustomerManagementComponent	/admin

10.2 Angular Service Registry

Service	Primary Responsibilities
AuthService	Manages JWT lifecycle in localStorage; provides currentUser() signal for role-based UI rendering.
CustomerService	Drives multi-step Quote Wizard; manages DocumentVault multipart/form-data uploads.
PolicyService	Fetches dashboard data, plan templates; handles policy assignment operations.
ClaimsService	Submits and retrieves claims; tracks claim status with RxJS observables.
InspectionService	Fetches inspection queue; submits reports; auto-refreshes on surveyor decisions.
UnderwriterService	Fetches verification queue; processes adjudication decisions; refreshes lists on success.
AdminService	Aggregates KPI data for dashboard; manages staff accounts and plan templates.
PaymentService	Luhn-validated payment form; calls payment endpoint; handles invoice PDF download.

10.3 Quote Wizard – 6-Step Flow

- Plan Selection: Customer selects Basic / Standard / Premium plan tile.
- Property Details: Enter property address, construction type, property value, and risk zone.

- Quote Calculation: CustomerService calls GET /api/quote/calculate; premium/coverage shown with animation.
- Account Confirmation: Anonymous users prompted to register; authenticated users proceed.
- Document Upload: Multi-file uploader with type/size validation and progress indicators.
- Submission: POST /api/quote/request; success state shows SubscriptionReference with live status tracker.

10.4 Admin Dashboard Analytics

- 5 KPI cards: Claims Approval Rate, Average Claim Amount, Average Settlement Time, Active Policy Ratio, Pending Inspections.
- Quick access panel: Recent Claims (last 10) and Pending Subscriptions (last 10).
- Pie chart: Claim status distribution using ng2-charts / Chart.js (optional).
- Bar chart: Monthly new policy subscriptions trend — 12 months (optional).
- Line chart: Monthly premium revenue trend — 12 months (optional).

10.5 UI Design Principles & Validation Rules

UI Feature	Specification
Design Language	Modern card-based layout with responsive sidebar; glassmorphism accent elements
Login Layout	Split-screen: 60% marketing content (hidden mobile) / 40% auth portal
State Signals	Angular Signals (isLoading, isSuccess, isError) for instant UI feedback
Status Badges	Green = Active/Approved/Settled; Amber = Pending/Inspecting; Red = Rejected
Analytics Charts	Pie (claim status), Bar (monthly policies), Line (revenue) via ng2-charts/Chart.js
Form Validation	Payment Luhn algorithm (client-side); file type/size; date validations
Responsive Design	Mobile-first; breakpoints at 768px (tablet) and 480px (phone)
Route Protection	CanActivate guard verifies JWT role before component initialization
Loading States	Skeleton loaders for tables; spinner overlays for form submissions

Chapter 11: Audit Log Dictionary & Invoice Generation

11.1 Audit Log Compliance Standards

The AuditActionLog is append-only by design — no entry is ever modified or deleted — providing a tamper-evident historical record for regulatory audit and forensic reconstruction. All timestamps are stored in UTC to eliminate timezone ambiguity. The admin can filter logs by date range, actor, and action type, and export to CSV/PDF.

11.2 Audit Action Type Dictionary

ActionType	Trigger	Fields Captured	Actor
PolicyVerification	Underwriter approves or rejects application	Status: Old to New	Underwriter
ClaimDecision	Underwriter approves or rejects claim	Status: Old to New	Underwriter
ClaimApprovalDeduction	Coverage deducted on approval	CoverageAmount: Before to After	System (Underwriter-triggered)
InspectionAssigned	Admin assigns surveyor to claim	SurveyorId: null to UserId	Administrator
InspectionReportSubmitted	Surveyor submits inspection report	Status: InProgress to Completed	Surveyor
UnderwriterAssignment	Admin assigns underwriter to policy	UnderwriterId: null to UserId	Administrator
DocumentReview	Staff sets document status	Status: Pending to Verified/Rejected	Underwriter
PolicyActivation	Payment confirmed; policy goes live	IsPaid: false to true; Status: Active	System (Customer-triggered)
StaffOnboarded	Admin creates staff account	New UserId record	Administrator
PlanTemplateCreated	Admin creates plan template	PlanId: created	Administrator
ClaimSettled	Settlement payment disbursed	Status: Approved to Settled	System
CustomerDeactivated	Admin soft-deletes account	IsActive: true to false	Administrator

11.3 Automated Invoice Generation (InvoiceService)

The InvoiceService uses iText/JasperReports to generate branded PDF invoices, invoked exclusively by the PaymentController after a confirmed payment. Generation and vault storage occur in a single synchronous operation within the payment transaction.

Section	Content	Data Source
Document Header	Company name, TransactionReference, PaymentDate	PaymentController parameters
Billed To	Customer full name and email	User.fullName, User.email

Policy Details	SelectedPlanName, TierName, CoverageAmount	PolicySubscription entity
Transaction Table	Description, Period (1 Month), Rate, Total	PolicySubscription.monthlyPremium
Footer	Page number, compliance disclaimer, issue date	System-generated

11.4 Invoice & Settlement Notice Storage Protocol

Generated PDF byte arrays are stored as new DocumentVault entries: DocumentType = Invoice or SettlementNotice, Status = Verified (no underwriter review required), RelatedEntityType = Policy or Claim. This ensures every activated policy and settled claim has an immediately accessible, immutable document available for customer download from their dashboard.

Chapter 12: Testing, QA & Non-Functional Requirements

12.1 Testing Strategy

Test Type	Framework	Scope	Owner
Unit Tests	JUnit + Mockito / Jest	PremiumService engine; boundary property values	Dev Team
Integration Tests	Spring Boot Test / Supertest	Payment to Invoice to Vault pipeline; rollback on failure	Dev Team
API Contract Tests	Postman / REST Client / Swagger	All 21 endpoints; happy path + error path validation	QA Team
Frontend Tests	Jasmine + Karma / Cypress	Loading states, form validation, role guard redirects	QA Team
Security Tests	OWASP ZAP / Manual	JWT bypass, RBAC escalation, SQL injection, file upload abuse	Security Team

12.2 Premium Engine Test Cases

Test ID	Input	Expected Output	Category
TC-01	PropVal=$100K, Basic, Base=$500	Premium=$550, Coverage=$66,000	Boundary – Min Value
TC-02	PropVal=$500K, Basic, Base=$500	Premium=$950, Coverage=$114,000	Boundary – High Value
TC-03	PropVal=$200K, Standard, Base=$500	Premium=$780, Coverage=$121,680	Normal – Standard Tier
TC-04	PropVal=$300K, Premium, Base=$500	Premium=$1,200, Coverage=$230,400	Normal – Premium Tier
TC-05	PropVal=$0, Basic, Base=$500	Validation error: Value must be > 0	Boundary – Zero Value
TC-06	PropVal=$100K, Invalid Plan	HTTP 400: Plan not found	Negative – Invalid Plan
TC-07	ClaimAmount > CoverageAmount	HTTP 400: Insufficient coverage balance	Deduction Gate
TC-08	ClaimAmount = CoverageAmount	Approved; CoverageAmount = 0	Deduction Gate – Edge

12.3 Non-Functional Requirements

NFR ID	Requirement	Target
NFR-PERF-01	Premium calculation API response time	< 200ms at 99th percentile (100 concurrent users)
NFR-PERF-02	Policy dashboard load time	< 1 second for up to 50 active policies

NFR-PERF-03	Invoice PDF generation time	< 3 seconds per document
NFR-PERF-04	Concurrent user support	100 simultaneous authenticated users
NFR-PERF-05	Audit log query (last 10,000 entries)	< 500ms via indexed SQL query
NFR-REL-01	System availability (excluding maintenance)	99.5% uptime per calendar month
NFR-REL-02	Data loss on unexpected shutdown	Zero loss for committed transactions
NFR-REL-03	Claim deduction atomicity	100% ACID; zero partial deductions permissible
NFR-REL-04	JWT validation failure handling	Graceful HTTP 401; no system crash or data exposure
NFR-SEC-01	BCrypt work factor	Minimum factor 12
NFR-SEC-02	TLS version	Minimum TLS 1.2; TLS 1.3 preferred
NFR-USA-01	Inline form validation feedback	Within 100ms of field blur events
NFR-USA-02	Status badge contrast ratio	WCAG 2.1 AA minimum 4.5:1

12.4 Usability & Accessibility

- Supports Evergreen browsers: Chrome 120+, Edge 120+, Firefox 121+, Safari 17+.
- Fully responsive: mobile breakpoints at 768px (tablet) and 480px (phone).
- Inline form validation feedback within 100ms of field blur events.
- Status badges maintain WCAG 2.1 AA contrast ratio (minimum 4.5:1).
- Skeleton loaders and progress indicators prevent perceived blank loading states.

Chapter 13: Deployment, Scalability & Future Roadmap

13.1 Deployment Architecture

Component	Hosting Option	Minimum Spec
Backend API	Tomcat / PM2 on Linux Server or Docker container	4 vCPU, 8 GB RAM, 100 GB SSD
Database	Managed DB (RDS, Cloud SQL) or self-hosted	8 vCPU, 16 GB RAM, 500 GB SSD (RAID)
Frontend (Angular)	Nginx Static Hosting or CDN (CloudFront, Vercel)	Standard CDN; 1 GB storage
SSL/TLS	Reverse proxy (Nginx / Caddy)	Wildcard SSL; minimum TLS 1.2

13.2 Environmental Configuration

Critical per-environment configuration managed via environment files and secrets manager:

- DB_CONNECTION_STRING — encrypted database credentials.
- JWT_SECRET_KEY — 256-bit symmetric key; unique per environment; never committed to source control.
- JWT_ISSUER / JWT_AUDIENCE — must match deployed domain URLs.
- CORS_ALLOWED_ORIGINS — exact Angular frontend URL(s) for the target environment.
- SMTP / EMAIL_API_KEY — transactional email service credentials.

13.3 Scalability Design

FireShield is designed for horizontal scalability. JWT stateless authentication means multiple API instances can run behind a load balancer without sticky sessions. Read-heavy operations (dashboard analytics, audit log queries) use read-only DB replicas to reduce primary load. The Angular build produces static artifacts distributable via CDN for global edge delivery.

13.4 Phase 2 & 3 Extensibility Roadmap

Phase	Feature	Technology	Business Driver
2	Cloud Object Storage for DocumentVault	AWS S3 / GCS	SQL storage relief; CDN delivery
2	Automated Email Notifications	SendGrid / SES API	Policy and claim status alerts
2	AI OCR Document Scanning	Google Document AI / Tesseract	Reduce manual review time by ~70%
2	Real-Time Push Notifications	WebSockets / SSE	Live status updates without polling
2	Live Payment Gateway	Razorpay / Stripe SDK	Real card processing (currently simulated)
2	Visual Analytics Charts	ng2-charts / Chart.js	Admin business intelligence dashboard

3	Risk Scoring Engine	ML model / Azure ML	Dynamic premium based on property risk
3	Fraud Detection Model	Anomaly Detection API	Auto-flag suspicious claim patterns
3	Mobile Application	React Native / Flutter	Native iOS and Android customer portals
3	Multi-Currency Support	Currency Exchange API	International market expansion
3	Geospatial Risk Mapping	Google Maps / Mapbox API	Location-based fire risk zone visualization

13.5 API Versioning & Migration Strategy

URL-based versioning (/api/v1/, /api/v2/) will be implemented as the system evolves. Version 1.0 endpoints remain frozen once v2.0 is released, with a minimum 6-month deprecation window. All new monetary schema columns must use DECIMAL(18,2); all new FK relationships must use cascade RESTRICT; all new audit-relevant entities must declare a corresponding ActionType in the Audit Log Dictionary.

Chapter 14: Glossary, Acronyms & Revision History

14.1 Glossary of Terms

Term	Definition
Coverage Amount	Remaining financial balance on a policy, decremented on each approved claim.
Deduction Gate	Business rule in claim adjudication blocking approvals that exceed the current CoverageAmount.
Document Gate	Business rule in policy verification blocking approvals unless all linked documents are Verified.
DocumentVault	Centralized binary document storage for both customer uploads and system-generated invoices.
Inspection	On-site property assessment conducted by a Surveyor to evaluate fire damage extent and loss value.
MRR	Monthly Recurring Revenue — sum of all active policy MonthlyPremium values.
Plan Template	Admin-configured insurance product (Basic/Standard/Premium) with base premium and coverage parameters.
PolicySubscription	The core entity representing a customer's insurance application and activated policy.
RBAC	Role-Based Access Control — restricts system access based on authenticated user roles.
Surveyor	Field staff role responsible for conducting property inspections and submitting damage reports.
Underwriter	Staff role responsible for policy risk assessment, document verification, and claim adjudication.

14.2 Acronyms

Acronym	Full Form
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
DTO	Data Transfer Object
FSMS	FireShield - Fire Insurance Management System
JWT	JSON Web Token (RFC 7519)
KPI	Key Performance Indicator
MRR	Monthly Recurring Revenue
ORM	Object-Relational Mapper
OWASP	Open Web Application Security Project

PII	Personally Identifiable Information
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
TLS	Transport Layer Security

14.3 Revision History

Version	Date	Author	Summary
0.1 Draft	2026-01-15	Technical Lead	Initial scope, stakeholders, functional requirements Ch. 1-4
0.5 Beta	2026-02-10	System Architect	Added API contracts, database schema, state machines
0.8 RC	2026-03-01	QA Lead + Architect	Security review; testing and NFR chapters added
1.0 Final	2026-03-18	Technical Team	Full review; glossary, roadmap finalized; approved for release

— End of Document —

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